
Autolume: A GAN-based No-Code AI Tool for Art Creation

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Abstract

Art creation through personally-trainable models, such as GANs, offers advantages such as creating unique aesthetics -without the need to verbally articulate them- and providing an intimate creative process through *small-data*. Yet, manipulating and navigating generative models can be technically challenging. To that end, we present Autolume, a no-coding generative AI system allowing artists to train, manipulate and navigate their own models.

1 Introduction

Recent advances in large-scale text-to-image diffusion models, such as DALL-E 3 by OpenAI [15] and Stable Diffusion by Stability AI [16], have significantly enhanced the ability to generate visuals from textual descriptions and led to the development of user-friendly tools that further democratize access to AI-driven art creation. Prior to that, small and personally-trainable models such as Generative Adversarial Networks (GANs) were more commonly used by AI artists, such as Helena Sarin, Memo Akten, Mario Klingemann, and Anna Ridler, among others.

Personal small models offer a distinctive value for artists over diffusion models which warrant continued support and attention from the AI community. Firstly, smaller models can be trained on smaller datasets, utilizing computing resources that are available to a wider range of artists. Working with smaller, personal or curated datasets in tandem with personally-trainable models such as GANs offers a highly controllable and intimate artistic workflow that fosters creative ownership and provides a sense of craftsmanship [6]. Secondly, they avoid perpetuating of biases associated with using large-scale models [17], which may be trained on datasets of artworks lacking proper attribution to creators. Thirdly, from an aesthetics point of view, models such as GANs provide a smoothly navigable and serendipitous latent space. Forthly, unlike large-scale text-based diffusion models, models such as GANs do not require the verbal articulation of tacit artistic intent [10], which is important for some artforms (e.g. Arabic calligraphy), where there is an inherent lack of vocabulary for describing the pieces, whether due to the nature of the artform or because they are not present in training datasets. Lastly, AI artists work with algorithms and data as materials for creation[9], and so by virtue of being faster to train or fine-tune, variations of smaller model can become easier to explore[9]. Smaller models also offer high levels of creative control as they are amenable to hacking [7], adapting, and crafting, e.g. through active divergence techniques such as network bending or blending [8].

2 Autolume: System Features

Latent spaces can also be challenging to navigate without guidance, which text prompting provides, and manipulating generative models can be challenging and often requires technical expertise, which many artists may lack. To make the artistic potential of personally-trainable generative AI models, particularly GANs, more accessible, and to expand the possibilities of working with them in an interactive real-time setting, we developed Autolume. Initially developed as a live VJing tool [14],

Autolume is a no-coding, user-friendly visual synthesizer leveraging the artistic potential of GANs. It simplifies the entire GAN workflow, from dataset preprocessing and model training to real-time latent space exploration, feature extraction, and output upscaling, making it accessible to non-technical creatives. Moreover, Autolume introduces interactive generation via OSC messaging, enabling audio-reactive and other forms of interactive art, thus broadening its creative possibilities.

Many no-code diffusion-based AI art tools have been introduced[2, 3, 4], but only a few exist for GAN-based art creation [5, 1]. Autolume is distinguished by being free and locally run, as well as providing real-time (semantic and OSC-based) latent space navigation and model mixing. Autolume offers both offline (pre-processing, post-processing) and real-time (latent space navigation, OSC messaging, semantic manipulation) modules. Autolume can be accessed on <https://www.metacreation.net/autolume>, and examples of artworks can be found on the website and Figure 1.

2.1 Offline Setup

The offline module enables users to train StyleGAN2 [13] models from scratch or fine-tune pre-trained models, using either image datasets or frames extracted from uploaded videos. It provides data preprocessing, including resizing and framing, and supports both square and non-square datasets by incorporating contributions to the StyleGAN2-ada network. To enhance training with smaller datasets, it offers Adaptive Discriminator Augmentation (ADA) [12] and Differentiable Augmentation (DiffAug) [18] methods, each with various augmentation pipelines. Users can adjust hyperparameters like batch size, learning rates, and gamma, and the training runs on the user’s local GPU. While achieving optimal results may require multiple iterations, a robust GPU, such NVIDIA GeForce RTX 3090, can yield meaningful outcomes in about 12 hours, with real-time progress monitoring available via a grid display of generated images.

The projection module in Autolume allows users to enter an image, text prompt, or both to find the closest latent vector in a trained model, as well as generating a video of the search process. The semantic feature extraction module uses GANSpace [11] to identify and extract interpretable feature directions in the latent space, with options for feature sparsity and the number of directions, both projection and feature extraction can be explored further in the real-time module.

2.2 Autolume-live: Real-time Navigation

The Autolume-live module enables artists to load models and work with them in real-time. It provides artists with diverse features for exploring the latent space and experimenting with the network’s parameters. It also enables artists to control the parameters via OSC protocol for audio-reactive works among others. This module also provides the ability to mix two trained models together by combining layers from each and observing the results in real-time. In addition, artists can add noise or apply custom transformations (e.g., translation, scaling, inverting) to chosen models’ layers.

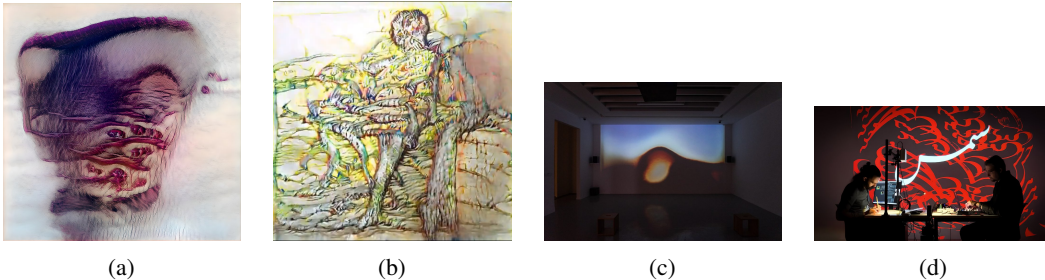


Figure 1: Works created with Autolume. (a) Dreamscape (b) Acedia (c) Mzton (d) Reprising Elements

3 Conclusion

We presented Autolume, a tool for art creation that adopts a model crafting and small-data mindsets. The goal of this project is to give artists the means to work in ways that maximize creative ownership and minimize ethical concerns associated with large-scale models. In the future, we will continue to develop Autolume and explore its potential through collaborations with artists.

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